

TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Chapter 1

CMS – Countermeasures Management Switch

1.1 Concept

The Countermeasures Management Switch (CMS) is a four-direction hat switch mounted on the control stick. It provides single-point control of the aircraft's defensive systems: the ALE-47 CMDS (Countermeasures Dispenser Set — the automatic chaff and flare dispensing system) and ECM (Electronic Countermeasures) systems, either external pods or internal avionics, depending on block configuration.

CMS actuation procedures are detailed in [Section 1.2](#). F-16 blocks and variant differences are covered in [Section ??](#).

For in-depth coverage of CMDS and ECM operation, see *TO 1F-16CMAM-34-1-1*, Dash-34, sections 2.7.1 and 2.7.2, respectively. Diagrams of the flight stick and throttle controls are provided in Dash-34 § 2.1.5. [Figure 1.1](#) shows the CMS location on the F-16 control stick.



Figure 1.1: F-16 Throttle and Flight Stick. Image by Falconpedia (falcon4.wikidot.com), via Wikimedia Commons (https://commons.wikimedia.org/wiki/File:F-16_Side_Stick_Controller.jpg), licensed under the Creative Commons Attribution-Share Alike 3.0 Unported (CC BY-SA 3.0) license.

1.2 CMS Actuation

The tables in this section cover CMS usage by system: CMDS (Manual, Automatic, and Semi-Automatic modes) and ECM integration with both external pods (ALQ-131/ALQ-184) and internal avionics (IDIAS).

CMS actuation is independent of the Master Mode currently selected.

Each table entry specifies:

- **Mode:** The operational context (Master Mode).
- **Direction:** CMS hat direction (Up, Down, Left, Right).
- **Action:** Press type (Short, Long, Long Hold).
- **Function:** What the CMS command activates or controls.
- **Effect / Nuance:** Resulting system behavior, including tactics and constraints.
- **Dash34:** Reference section in the Dash-34 manual.
- **Training:** Recommended BMS training mission(s) (see BMS Training Manual 4.38.1 for descriptions).

1.2.1 CMDS

The ALE-47 CMDS provides three operational modes: Manual (MAN), Automatic (AUTO), and Semi-Automatic (SEMI), each described below.

The CMS is the control interface for program execution and consent authority in all three modes.

1.2.1.1 Manual Mode

In Manual (MAN) mode, the pilot executes countermeasure programs directly via CMS. Each CMS direction (**CMS Up** or **CMS Left**) executes a specific program.

Table 1.1: CMS Actuation with CMDS Manual Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS MAN	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once per press. No threat sensing; no automatic triggering. Overrides any AUTO dispensing, if running.	2.7.2.2	18, 28
	Left	Short	Execute Program 6	Flare-only program. PRGM knob position has no effect. Independent of threat sensing.	2.7.2.2	19

1.2.1.2 Automatic Mode

In Automatic (AUTO) mode, the ALE-47 CMDS dispenses the selected program (1–4) *continuously* in response to threats detected by the Radar Warning Receiver (RWR).

Pilot consent is given *once* by pressing **CMS Down**; dispensing continues until **CMS Right** is pressed or expendables are exhausted.

Table 1.2: CMS Actuation with CMDS Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, interrupting any ongoing AUTO dispensing. After execution, AUTO resumes if the threat persists.	2.7.2.2	18, 28
CMDS AUTO	Left	Short	Execute Program 6	Manual flare-only program, overrides AUTO. After execution, AUTO resumes.	2.7.2.2	19
CMDS AUTO	Down	Short	Give Consent; Enable AUTO Dispensing	Grants consent for AUTO dispensing. RWR-detected threats trigger automatic execution of the program selected via the CMDS panel PRGM knob. Dispensing continues until the threat clears or CMS Right is pressed. Consent state persists if the pilot switches to MAN mode; returning to AUTO resumes auto-dispensing if a threat is detected.	2.7.2.1	18, 28
CMDS AUTO	Right	Short	Cancel Consent; Disable AUTO	Removes CMDS consent and places the ALE-47 in Standby. Automatic dispensing halts immediately. Pilot must re-issue CMS Down to resume AUTO operation..	2.7.2.1	18, 28

1.2.1.3 Semi-Automatic Mode

In Semi-Automatic (SEMI) mode, the ALE-47 CMDS dispenses *one program* per RWR-detected threat, with pilot consent required for *each event*.

When a threat is detected, the CMDS control unit displays DISPENSE RDY and the system sounds a COUNTER voice message.

Table 1.3: CMS Actuation with CMDS Semi-Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS SEMI	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, independent of RWR threat state. After execution, CMDS resumes SEMI monitoring.	2.7.2.2	18, 28
CMDS SEMI	Left	Short	Execute Program 6	Manual flare-only program, independent of SEMI threat detection. After execution, CMDS resumes SEMI monitoring.	2.7.2.2	19
CMDS SEMI	Down	Short	Give Consent; Dispense One Program	Executes one instance of the selected program in response to a COUNTER prompt. If the threat persists or a new threat appears, CMDS issues COUNTER again. Consent state is tracked; switching to AUTO while consent is active resumes auto-dispensing immediately on the next threat.	2.7.2.1	18, 28

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Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMD5 SEMI	Right	Short	Cancel Consent; Return to Standby	Removes SEMI consent. CMD5 halts monitoring and returns to Standby. COUNTER messages cease.	2.7.2.1	18, 28

1.2.2 ECM

External ECM pods (ALQ-131, ALQ-184) and internal IDIAS (Improved Defensive Internal Avionic System) provide active jamming across radar frequency bands. **CMS Down** grants consent to external pods. **CMS Left** cycles operational modes for internal IDIAS.

1.2.2.1 External ECM Pod

CMS Down grants consent for ECM pod transmission. The XMIT switch on the ECM control panel sets the transmission mode (XMIT 1: AUTO Avionics Priority; XMIT 2: AUTO ECM Priority; XMIT 3: Continuous Jam). The ECM Enable light on the miscellaneous panel reflects consent state. Both systems are inhibited when the pilot moves the RF switch from NORM, or when the landing gear handle is down.

Table 1.4: CMS Actuation with External ECM Pod

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
ECM Pod	Down	Short	Grant ECM Consent	CMS Down illuminates the ECM Enable light. The pod transmits in the mode set by the XMIT switch. Transmission continues until CMS Right is pressed or the RF switch is moved from NORM.	2.7.4.2.5	28
ECM Pod	Right	Short	Remove ECM Consent	CMS Right removes consent and places the pod in Standby. The ECM Enable light extinguishes. Transmission does not resume until the pilot re-issues CMS Down.	2.7.4.2.5	28

1.2.2.2 Internal ECM (IDIAS)

IDIAS provides three operational modes controlled via **CMS Left**: Standby (STBY), Avionics Priority (AVNC), and ECM Priority (ECM). Operational modes are available only when the XMTR button is set to OPER. The active mode is reflected on the HUD, FCR, RWR, and TFR pages.

Table 1.5: CMS Actuation with Internal ECM (IDIAS)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
IDIAS ECM	Left	Short	Cycle ECM Operational Mode	Each short press of CMS Left advances the IDIAS mode: STBY → AVNC (Avionics Priority) → ECM (ECM Priority) → AVNC → ECM.	2.7.4.1.2	28
IDIAS ECM	Right	Short	Remove ECM Consent	CMS Right sets IDIAS to STBY, removing consent. AVNC or ECM modes are not restored until CMS Left is re-applied.	2.7.4.1.2	28

1.2.3 Consent and Constraints

When equipped with an external ECM pod, **CMS Down** grants consent to both CMDS (in AUTO or SEMI mode) and the external ECM pod *simultaneously*.

Table 1.6: CMS Interaction with CMDS and ECM (Consent and Constraints)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO SEMI + ECM Pod	Down	Short	Joint Consent (CMDS + ECM)	A <i>single</i> CMS Down press grants consent to <i>both</i> the ALE-47 CMDS (AUTO or SEMI mode) and the external ECM pod.	2.7.2.1, 2.7.4.2.5	18, 28

1.2.4 Operational Notes

State Tracking: In AUTO and SEMI modes, the CMDS retains consent state when the pilot switches to MAN mode. If the pilot re-engages AUTO without pressing **CMS Down**, the CMDS begins dispensing immediately on the next detected threat.

Bingo Quantity Behavior: If expendables fall to or below the bingo quantity, the CMDS requests consent and continues dispensing. The VMU issues a **LOW** message when an expendable reaches the bingo quantity, and an **OUT** message when an expendable is depleted. Dispensing does not stop automatically. The pilot can press **CMS Right** to inhibit AUTO dispensing, or switch to MAN mode to control expenditure directly.

RF Switch Override: When ECM consent is active, both the external ECM pod and internal IDIAS are placed in Standby if the RF switch is moved from NORM to QUIET or SILENT. Returning the RF switch to NORM does *not* restore consent. The pilot must re-issue **CMS Down** to resume ECM transmission.

IDIAS vs. External ECM Pod: The joint consent described in [Table 1.6](#) applies to external ECM pod configurations *only*. On IDIAS-equipped aircraft, **CMS Left** is used to cycle ECM modes; **CMS Down** has no ECM function on these variants.

Ground Safety: On the ground, ECM consent state is Standby. If the pilot holds **CMS Down** on the ground, the ECM pod radiates until the pilot releases **CMS Down**. Ground personnel must be clear of the radiation area if, for any reason, ECM must be enabled on the ground.